Designing for Recovery: Affective Design Patterns After Failure in Learning Software

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Abstract

Learning software is full of failure: wrong answers, failed tests, broken builds. We reviewed 41 learning applications and interviewed 12 designers to collect patterns for the moment after a learner fails. We describe nine patterns, from "name the next smallest step" to "make the failure cheap to retry", and discuss their affective rationale with reference to broaden-and-build theory and emotional design.

1 Introduction

Most affective design guidance concerns delight. Learning software needs guidance for the opposite moment, when the learner has just been told they are wrong.

2 Method

We reviewed 41 learning applications (programming, language, mathematics) by deliberately failing tasks and recording the response. We then interviewed 12 designers who had worked on such applications about decisions behind failure screens.

3 Patterns

Pattern	Seen in (of 41)	Rationale
Name the next smallest step	14	narrows attention to something achievable
Make retry cheap	27	lowers the cost of trying again
Separate the person from the attempt	9	avoids "you are wrong" framing
Show partial credit	12	makes progress visible
Delay the red	5	lets the learner read before seeing a failure color
Normalize the difficulty	8	"most people miss this"
Offer an exit	6	a different task, not just a hint
Preserve work	22	never lose what the learner typed
Invite reflection later	3	return to the failure once it is solved

4 Discussion

Designers described tension between honesty and comfort. "Normalize the difficulty" works only when it is true. Several designers had removed celebratory animations after failures were resolved because learners found them patronizing.

5 Limitations

A review by one designer-researcher, and an interview sample recruited through professional networks.

References

[1] Norman, D. A. Emotional Design: Why We Love (or Hate) Everyday Things. Basic Books, 2004.

[2] Fredrickson, B. L. The role of positive emotions in positive psychology. American Psychologist 56, 3 (2001), 218-226.